Vision Language Models Lab Worksheet

Syeda Maham Mohsin

Student ID: k230594

Step 1 CLIP results

Labels and prefix tried	Top label and score	Surprised Y or N
cat, dog, car, sofa; a photo of	sofa - 68.1%	Y
one cat, two cats, three cats; a photo of	two cats - 74.0%	N
cat, dog, car, sofa; empty prefix	sofa - 73.8%	Y
cat, dog, car, sofa; a drawing of	sofa - 74.2%	Y
cat, dog, car, sofa, a spaceship; a photo of	sofa - 68.0% a spaceship - 0.1%	Y

Reflection: Sofa ranked above cat even though two cats are visible. Changing the prefix changed the text embeddings and therefore the relative scores. Softmax distributes scores across the supplied labels, so even the unrelated spaceship label received a small nonzero score. The surprise entries reflect these observations.

Step 3 Failures found

Category	Model answer and true answer	My explanation	One change tried and result
Counting	7 circles: answered 5. 12: answered 9. 20: answered 15.	The model may estimate from visual features instead of counting each object explicitly.	Kept the prompt fixed and varied the circle count: 3 -> 3, 7 -> 5, 12 -> 9, 20 -> 15.
Left and right	Square on left: answered square. Circle on left: answered circle. Both correct.	No spatial failure was observed with these two simple shapes.	Swapped the shapes while keeping the prompt fixed. The answer changed correctly.
Text OCR	Size 40: read the message correctly. Size 14: gave a generic description instead of transcribing it.	Small letters may lose detail during image processing, weakening the visual evidence for exact reading.	Reduced font size from 40 to 14 at 0 degrees; transcription failed. Separately, size 40 at 20 degrees remained correct.
Hallucination	Answered that no dog was present. True answer: there is no dog.	The model rejected the false premise in this test; no dog hallucination was observed.	Changed the question to Is there a dog in this picture? It again correctly answered no.
Own photo	Described a classroom; answered White for the children's uniforms. White shirts are visible.	The broad description and shirt colour match the photo. The specific claim about Chinese characters is not verified.	Changed the broad description prompt to a specific uniform colour question. Answer: White. CLIP separately chose a whiteboard at 52.9% from limited labels.

My best failure

SmolVLM undercounted circles as the number of objects increased. It counted three circles correctly, but answered only 15 when the image contained 20 circles.

The test I ran

I generated images containing 3, 7, 12, and 20 circles using make_dots(n). I asked the same question for each image: "How many circles are in this image? Answer with a number." I kept the image size, generation method, seed setting, and prompt fixed while changing n.

True circle count	SmolVLM answer	Undercount
3	3	0
7	5	2
12	9	3
20	15	5

My explanation

My explanation is that the model relies on an overall visual representation rather than an explicit object-by-object counting procedure. As more circles appear, it may merge or miss some objects and produce an estimate. This would explain why it answered correctly for the simplest image but underestimated the larger counts.

What the result shows

The absolute counting error increased from 0 to 2, 3, and 5 across the four images. This supports my explanation that counting becomes harder as the number of objects increases. It does not prove the internal cause: changing n also changes the arrangement and density of the circles, and I tested only one generated image for each count.

When I would use CLIP

I would use CLIP when I need to quickly classify or retrieve images using a known set of text labels, rather than generate an open-ended description or answer.